Lecture 2 - Pollution, Climate Resilience and the Circular Economy

Lecturer notes, slide by slide

With activities, student questions and model answers

Deck	ASM_Lecture 2_Pollution Climate Resilience and Circular Economy_revised.pptx
Slides	35 - numbering matches the revised PowerPoint
Lecturer	Jenan Hussein
Revision	5 October 2026; content and visual approach adapted from approved Lecture 1
Teaching use	Read the summary first, then keep the slide cards beside you. Say gives the explanation; Ask and Tip support delivery.

What changed

Seven focused objectives; clearer links to business decisions; the original six-part sequence, cases and activities; corrected qualifications where benefits or rules were too absolute. Full teaching notes are also embedded in the PowerPoint.

Seminar connection

Seminar 2 (Sustainable Development Goals, European Green Deal). The circular economy part prepares students for the group project, where they must design a business that uses circular principles.

1. The lecture at a glance

The big idea

Companies face pollution costs, climate disruption and loss of material value. The first half links urban problems to company operations, then compares nature-based adaptation on business sites with city-scale examples. The second half explains circular strategies and resource loops. Keep returning to four questions: what problem is being solved, who pays, who maintains the solution, and how will its performance be measured?

Learning outcomes

1. Explain how pollution and emission scopes affect business decisions.
2. Identify climate risks to operations, workers and suppliers.
3. Compare nature-based and engineered adaptation options.
4. Distinguish linear and circular business models.
5. Use the R-ladder to preserve product and material value.
6. Evaluate circular practices in company and city cases.
7. Design a resource loop with a payer, owner and measure of success.

Words to explain early

Term	Plain-language meaning
Scope 1, 2 and 3	Direct emissions from owned or controlled sources; purchased energy emissions; other value-chain emissions.
Urban resilience	The capacity of people, businesses and systems to cope with stresses and shocks.
Nature-based solution	Use of ecosystems or natural processes to address a problem while supporting people and biodiversity.
Linear economy	Materials are extracted, made into products, used and discarded.
Circular economy	An approach that aims to keep products and materials useful, reduce waste and regenerate nature.
R-ladder	Strategies such as refuse, reduce, reuse, repair and remanufacture generally retain more product value than recycling.
Doughnut Economics	A framework that considers a social foundation alongside ecological limits; used in the Amsterdam case.

Timing plan

These are the original suggested block allocations, retained for planning. Adjust them to your actual teaching slot. Individual slide timings are approximate; do not add them to the block totals.

Block	Slides	Minutes
Welcome, objectives, agenda	1-3	8
Part 1 - Pollution in the urban environment	4-8	25
Part 2 - Climate change and urban resilience	9-11	15
Part 3 - Nature-based adaptation for business	12-15	20
Prague case: heat islands + activity 'Cooling one street'	16-18	25
Break	19	15-20
Part 4 - From linear to circular	20-24	25
Part 5 - Circular economy in business practice	25-27	15
Part 6 - Green Infrastructure and urban circularity	28-30	15
Activity: redesign for circularity	31	20
Key messages, reading, questions	32-34	10

Before class

Bring one everyday object (a coffee cup, a phone charger, a T-shirt) - you can use it on slides 21 and 31 to make the linear/circular idea concrete. Optional: check the current air quality index for Prague on the day (for slide 5). Remind yourself which Prague site each group chose in Lecture 1 - slide 18 works best when they use their own site.

Use the revised deck named on this cover. Confirm the timetable, seminar deadline and assessment details with the course organiser. Distinguish classroom assumptions from observed results.

2. Slide-by-slide notes

Slide numbers and titles match the revised deck. These are suggested words and prompts, not a script to read verbatim.

01 Applied Sustainability in Business and Management (1 min)

Say: Welcome back. Remind students: last time we saw why sustainability matters and what Green Infrastructure is. Today we look at the damage cities cause and two ways to fix it.

02 Learning Objectives (3 min)

Say: Read the seven objectives. Connect each to a decision a manager needs to make. The first half introduces the tools and business problem; the second half applies them to the cases and activities. Students will use the same tools for their Prague semester site.

03 Today's Lecture Agenda (2 min)

Say: Show the six parts. Say there are two short activities today - one before the break and one at the end.

04 Part 1 - Pollution in the Urban Environment (1 min)

Say: Announce the part.

05 Air Pollution (5 min)

Say: The photo shows smog over a big city. The main sources in a city are cars, heating and industry. Air pollution causes breathing and heart disease, which costs money in sick days and healthcare. For a manager the point is: this is a cost, not only a health topic.

Ask: Where does Prague's winter smog come from? (Mostly heating and traffic.)

06 Water Pollution (4 min)

Say: Rainwater running off streets carries oil, metals and tiny plastics into the river. In heavy rain, old sewers overflow untreated. Why should managers care? Because water quality decides permits and reputation, and because a clean river creates tourism and property value.

07 Waste Generation (4 min)

Say: Cities concentrate people, so they concentrate waste. Landfills leak methane (a strong greenhouse gas) into the air and pollutants into the ground. Introduce the waste hierarchy: prevent, reuse, recycle, recover, dispose - the best option is at the top. Say this is the bridge to the second half of the lecture.

08 Carbon Emissions and Business (6 min)

Say: Scope 1 covers direct emissions from sources the company owns or controls, including fuel use, industrial processes and refrigerant leaks. Scope 2 covers purchased electricity, heat, steam and cooling. Scope 3 covers other value-chain emissions, from suppliers to product use and disposal. Its importance varies by business.

Ask: Which scope is hardest for this particular company to reduce, and what purchasing or design decision could help?

09 Part 2 - Climate Change and Urban Resilience (1 min)

Say: Announce the part: from pollution we move to climate.

10 How Climate Change Hits Cities (5 min)

Say: Three columns: heat (heat waves, hotter city centres, health risks), water (flash floods, river floods, drought, old drains that cannot cope) and economy (broken infrastructure, closed businesses, higher insurance). Ask students which of these they have seen in Prague. Most will remember a flood or a heat wave.

11 What is Urban Resilience? (5 min)

Say: Resilience means a business and its surrounding systems can keep essential functions working during stress or recover after a shock. Use redundancy, flexibility and nature as a buffer. Wetlands and trees can help manage heat and runoff, alongside drainage and engineered protection; none is unlimited.

12 Part 3 - Nature-Based Adaptation for Business (1 min)

Say: Move from city-level climate risks to what a company can do on its own buildings and sites, then show how those measures connect to city infrastructure.

13 Nature-Based Adaptation on Company Sites (5 min)

Say: Use an office, shop or warehouse as the example. A roof can retain rain, trees can shade entrances, and a rain garden can slow runoff. The manager must check the building structure, ground conditions, drainage and maintenance budget. These measures can reduce exposure, but their capacity is limited and overflow routes still matter.

Ask: Who owns the problem, who pays for installation, and who maintains the system?

Source references: see the source list at the end of this guide.

14 Working with Water, not Against It (3 min)

Say: Benthemplein remains the city-scale case. It shows how one space serves sport in dry weather and water storage during storms. A business can apply that principle to a courtyard or car park if an engineer confirms capacity, safe access and overflow routes. This is a design analogy, not a claim that any car park can flood safely.

15 Grey and Green: a Manager's Comparison (5 min)

Say: Neither category is automatically cheaper or better. Compare the same flood or heat problem over the same period. Include installation, maintenance, performance limits and wider benefits. A combined system may provide the strongest case.

Ask: Which benefit does the company receive, which goes to the public, and how should that affect cost sharing?

Source references: see the source list at the end of this guide.

16 Prague Case Study - Heat Islands (1 min)

Say: Announce the case.

17 Heat Islands in Prague Neighborhoods (5 min)

Say: Use the Prague case to compare paved and shaded places, then relate heat to staff comfort, cooling demand and opening hours. A measured temperature difference depends on time, weather, shade and the sensor position. The city plans provide context; students should verify the current local project and site data before proposing an investment.

18 Classroom Activity: Cooling One Street (15 min)

Say: Pairs choose a familiar Prague street or square and propose three nature-based measures. For each, name the payer, the maintenance owner, the business or community benefit, and one measure of success. Keep the 15-minute structure in section 3.

19 Break (15-20 min)

Say: Say that after the break we change topic: from pollution and climate to materials and waste - the circular economy.

20 Part 4 - From Linear to Circular (1 min)

Say: Announce the part.

21 The Linear Economy (5 min)

Say: Take, make, waste. Hold up an everyday object and trace it: raw material, factory, shop, short use, bin. The linear economy assumes endless resources and endless space for waste - both are false. Value is created once and then destroyed.

22 The Circular Economy (5 min)

Say: Circularity is a design ambition to keep materials and products useful and regenerate nature. Real systems still have losses and need energy. Use the forest analogy to explain loops, then ask what collection, repair or quality-control system a company would need.

23 Circular Strategies (the R-Ladder) (6 min)

Say: The ladder ranks strategies. At the top: refuse, rethink, reduce, reuse, repair - these keep almost all the value. In the middle: refurbish, remanufacture, repurpose. At the bottom: recycle and recover energy. The higher you act, the more value you keep. Ask students where 'buying a second-hand phone' sits (reuse - high) and where 'putting a bottle in the yellow bin' sits (recycle - low).

24 Reuse, Repair and Then Recycling (3 min)

Say: The PET-bale image shows material recovery. Recycling remains useful, but viable reuse or repair can preserve the product as well as its material. Avoid claiming a fixed loss of embedded value across all materials. Ask students to explain which option works for this product and context.

25 Part 5 - Circular Economy in Business Practice (1 min)

Say: Announce the part: two real examples, one company and one city.

26 Case Study: LEGO Circular Design (6 min)

Say: LEGO illustrates durable design and repeated use. It also tests alternative materials and uses both physical recycled or renewable inputs and mass-balance sourcing. Those are different claims: a sourcing allocation does not establish the physical renewable content of an individual brick. Replay supports reuse where the programme operates.

Ask: Is durability itself a circular strategy? Yes, when products remain useful and replace the need for new ones.

Source references: see the source list at the end of this guide.

27 Case Study: Amsterdam Circular Strategy (5 min)

Say: A whole city can be circular. Amsterdam wants to halve its use of new raw materials by 2030. It has rules for circular construction and demolition, closes food and organic waste loops locally, and uses Doughnut Economics as its guiding picture. Lesson: circularity works best when city policy and business models push in the same direction.

Source references: see the source list at the end of this guide.

28 Part 6 - Green Infrastructure and Urban Circularity (1 min)

Say: Announce the part: how GI and the circular economy meet.

29 Resource Loops for Companies and Cities (5 min)

Say: Follow each loop from an input to a useful output. A company can collect rainwater, buy local produce or send suitable separated food waste for composting. Each loop needs a contract or owner, a quality requirement and a measurable benefit. Reused water and compost must meet the applicable quality and use requirements.

30 Urban Farming as Circular Infrastructure (4 min)

Say: The photo is a community garden in New Orleans, used to illustrate food, compost, rainwater and community connections. For Prague, ask students to identify a real local garden and verify its activities. A café or employer could be a buyer, contributor or maintenance partner; these are ideas to test, not existing contracts.

31 Classroom Activity: Redesign for Circularity (20 min)

Say: Groups choose a product or urban space and redesign its material flows. Keep the original 20-minute activity plan: 12 minutes to design and 8 minutes to share. They should identify losses, use one suitable local recovered input, and explain how the idea could restore nature. The pitch must include a payer and an operating owner.

32 Key Messages (4 min)

Say: Work through the eight messages. Ask students to connect each to a decision, example or tool from this lecture. Distinguish potential benefits from measured results.

33 Recommended Reading (2 min)

Say: Point to the Ellen MacArthur Foundation report - it is short and free online. Say the next lecture is about sustainable business models and corporate strategy.

34 Any questions? (5 min)

Say: Take questions. If none, use a question from section 5.

35 Image credits

Say: No need to show. Keep it in the file.

3. How to run the class activities

Cooling One Street (slide 18, 15 minutes)

1. Pairs choose a street or square in Prague they both know (or the street of their project site).

2. 8 minutes: propose three nature-based measures to cool it (trees, green facade, water element, permeable paving, shade structure with plants).

3. For each measure write: who would pay and maintain it (city, shop owners, building owners, a mix) and who benefits (residents, shops, the city).

4. 7 minutes: three or four pairs present. Ask the room: which measure gives the most benefit per crown?

Watch for: Students often propose measures without thinking about the payer. Push them: if the shop owners benefit from cooler customers, should they pay part?

Redesign for Circularity (slide 31, 20 minutes)

1. Groups of three or four. Each chooses one everyday product (coffee cup, sneakers, office chair) or one urban space (a school yard, a car park, a market).

2. 12 minutes: redesign it to meet the three rules on the slide. They draw a simple loop diagram: where materials come from, where they go, what comes back.

3. 8 minutes: one-minute pitch per group. Time them strictly - it is good practice for the final seminar presentation.

4. Close by connecting to the R-ladder: which rung did each group use most?

Watch for: Most groups jump to 'recycle'. Ask them to move up the ladder: can it be reused, repaired, or not needed at all?

Business decision prompt

For each proposal, ask students to name the payer, operating owner, affected stakeholders and one measurable outcome. Use assumptions openly when data is unavailable.

4. If a student asks

Is recycling bad then?

Recycling is useful for materials that cannot stay in higher-value loops. Where viable, prevention, reuse and repair retain more of a product and its manufacture. The best option depends on the product and conditions.

Can a company really control Scope 3?

Not directly, but it can choose suppliers, design products that use less energy, and help customers recycle. Big companies now ask their suppliers for emission data - so Scope 3 for a big company becomes Scope 1 for a small one.

Is a wetland as safe as a concrete flood wall?

The answer depends on the hazard, site, design capacity and maintenance. Nature-based measures can complement engineered protection, but neither a wetland nor a wall is unlimited. Compare a combined system against the same required service level.

Does the circular economy make things more expensive?

Sometimes at the start. But it also saves raw materials, creates repair jobs and protects a company from price shocks. LEGO's durable bricks are a profitable business, not a charity.

5. Questions with model answers

Use these for discussion, homework or test preparation. Model answers are short examples; other well-supported answers can also be valid.

1. Name three sources of urban air pollution and two ways it costs money.

Recall

Model answer: Sources: road traffic, heating and energy production, industry and construction. Costs: healthcare and sick days, lost productivity (also lower attractiveness for tourism and talent).

2. Explain Scope 1, 2 and 3 emissions. Which is usually the largest for a clothing brand, and why?

Understand

Model answer: Scope 1 covers direct emissions from sources owned or controlled by the company, including fuel combustion, processes and leaks. Scope 2 covers purchased electricity, heat, steam and cooling. Scope 3 covers other value-chain emissions. For a clothing brand, material production and customer use can make Scope 3 a large share; verify the inventory rather than assuming a fixed share.

3. What is the waste hierarchy? Write it in the right order.

Recall

Model answer: Prevent > Reuse > Recycle > Recover (energy) > Dispose. The best option is at the top.

4. Define urban resilience and explain the three principles for managers with one example each.

Understand

Model answer: Urban resilience means people, businesses and systems can cope with chronic stresses and sudden shocks. Redundancy provides an alternative critical service, flexibility adapts how an asset is used, and nature can buffer heat or runoff alongside engineered protection.

5. Match each problem to a nature-based tool and explain how it works: (a) hot street, (b) sewer overflow in storms, (c) high cooling bills in an office.

Apply

Model answer: (a) Street trees - shade and evaporation cool the air. (b) Rain gardens or a water square - they slow and store runoff so sewers are not overloaded. (c) A green roof - insulation reduces heat entering the building.

6. Explain the difference between a linear and a circular economy in two sentences each.

Recall

Model answer: Linear systems extract, make, use and discard materials. Circular approaches aim to retain useful products and materials through design, maintenance, reuse, repair and remanufacturing, then recovery where needed, while accounting for real losses and regeneration.

7. Put these actions on the R-ladder from highest to lowest value: recycling a bottle, repairing a phone, refusing a plastic bag, remanufacturing an engine.

Understand

Model answer: Refusing a plastic bag (refuse) > repairing a phone (repair) > remanufacturing an engine (remanufacture) > recycling a bottle (recycle).

8. Why is recycling called 'the last resort' in the circular economy?

Understand

Model answer: Recycling can recover materials but often loses product form and requires additional processing. Prefer viable prevention, reuse and repair where these preserve more value. Recycling remains useful for materials that cannot stay in a higher-value loop.

9. Give two things LEGO does that are circular and explain which R-strategy each one uses.

Apply

Model answer: Durable bricks support repeated use and resale. Replay supports reuse through donation where available. Material research and sourcing address inputs, but mass-balance allocation must be distinguished from the physical recycled or renewable content of an individual brick.

10. Describe three resource loops that Green Infrastructure can close in a city.

Recall

Model answer: Water: rainwater stored in green areas and treated greywater used for irrigation. Food: urban farms and community gardens grow food where it is eaten. Soil: compost from city bio-waste builds soil that stores water and carbon.

11. Choose a product you used today. Name the stage where most of its waste is created and propose one change that moves it up the R-ladder.

Apply

Model answer: Any reasoned answer. Example - take-away coffee cup: waste at the use stage (one use, then bin). Change: a deposit-return reusable cup system (reuse), or refusing the cup and using your own (refuse).

Handout version: questions only

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2. Explain Scope 1, 2 and 3 emissions. Which is usually the largest for a clothing brand, and why?

3. What is the waste hierarchy? Write it in the right order.

4. Define urban resilience and explain the three principles for managers with one example each.

5. Match each problem to a nature-based tool and explain how it works: (a) hot street, (b) sewer overflow in storms, (c) high cooling bills in an office.

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7. Put these actions on the R-ladder from highest to lowest value: recycling a bottle, repairing a phone, refusing a plastic bag, remanufacturing an engine.

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Source list for revised passages

Checked 5 October 2026. These sources support the revised passages; the original recommended reading and image credits remain in the deck. Company material is identified as company-reported evidence.

1. www.epa.gov - slides 13, 15

https://www.epa.gov/green-infrastructure/economic-benefits-green-infrastructure

2. www.lego.com - slides 26

https://www.lego.com/en-us/sustainability/materials

3. www.lego.com - slides 26

https://www.lego.com/en-us/sustainability/mass-balance

4. www.amsterdam.nl - slides 27

https://www.amsterdam.nl/en/policy/circular-economy/